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Terrain machine learning

Viktor Wiberg

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Viktor Wiberg
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Department of Physics
Umeå University
SE-907 36 Umeå
Sweden
Telephone: +46 (0)90-786 500

Cover: Simulated trajectories coloured by value function estimate.

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Abstract

The use of heavy vehicles in rough terrain is vital in the industry but has negative implications for the climate and ecosystem. In addition, the demand for improved efficiency underscores the need to enhance these vehicles' navigation capabilities. Navigating rough terrain presents distinct challenges, including deformable soil, surface roughness, and spatial and temporal terrain variability. Focusing on forestry, this thesis aims to improve navigation using machine learning and physics simulations. Without considering the vehicle-terrain dynamics, methods for navigation can result in unsafe or unnecessarily challenging situations. Specifically, we address route planning, control for autonomous vehicles, and soil deformations. We simulate soil using the discrete element method and vehicles using multibody dynamics.

To enhance route planning, we train a predictor model that uses a height map of the terrain to predict measures of *traversability*. The model has a directional dependency, couples geometric terrain features with vehicle design and dynamics, and allows for swift evaluations over large areas. The proposed method facilitates detailed route planning, using multiple objectives to yield efficient solutions.

We address autonomy in rough terrain navigation by training a controller through deep reinforcement learning. The control policy uses a local height map for perception to plan and control a forwarder with actively articulated suspensions. The controller adapts to overcome various obstacles and demonstrates skilled driving in rough terrain.

Extending beyond simulation, we address the simulation-to-reality gap of vehicles with complex hydraulic drivelines through system identification and domain randomization. The results show that having an accurate model of the actuators, modelling system delays, and preventing bang-bang control yields successful transfer. Controllers that train in simulation and transfer to reality are a step toward autonomous vehicles.

While the previously mentioned studies assume rigid terrain, we also answer if the discrete element method can capture large soil deformations due to heavy traffic. The results show that the discrete element method can represent a wide variety of natural soil and that the resulting rut depths agree well with empirical models and experimental data, including multipass scenarios.

Publications

This thesis is based on the following papers:

I Discrete element modelling of large soil deformations under heavy vehicles

V. Wiberg, M. Servin, T. Nordfjell

Journal of Terramechanics **93**, 11–21 (2021).

II Learning multiobjective rough terrain traversability

E. Wallin, V. Wiberg, F. Vesterlund, J. Holmgren, H.J. Persson, M. Servin

Journal of Terramechanics **102**, 17–26 (2022).

III Control of rough terrain vehicles using deep reinforcement learning

V. Wiberg, E. Wallin, T. Nordfjell, M. Servin

IEEE Robotics and Automation Letters **7**(1), 390–397 (2021).

IV Sim-to-real transfer of active suspension control using deep reinforcement learning

V. Wiberg, E. Wallin, A. Fälldin, T. Semberg, M. Rossander, E. Wadbro, M. Servin

Manuscript, (2023).

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Chapter 1

Introduction

Navigation capability in rough terrain refers to the ability of a vehicle to move through an environment that may contain obstacles, deformable soil, uneven terrain, slopes, and other natural features. However, accurately predicting vehicle behaviour in such situations is challenging due to the complex vehicle–terrain interaction, which depends on the terrain, soil properties, the vehicle, and its operation. A lack of accurate prediction limits the fidelity of operational planning and control, often resulting in the use of simplified kinematic models that neglect the actual dynamics of the vehicle. Without considering the dynamics, navigation strategies may lead to inefficient, unsafe, or soil-disturbing traversal. To address these challenges, physics simulations can be used to capture the vehicle–terrain dynamics and enable controlled and repeatable data collection. Combining simulators with machine learning has the potential to train advanced controllers for heavy vehicles and enhance tools for planning.

In this thesis, we develop methods to improve navigation capabilities for vehicles in rough terrain using physics simulations and machine learning. As we rely on simulations to model real-world interactions, we should be aware of when models are valid. In particular, we study the validity of soil deformations due to vehicles using the discrete element method (DEM). However, the high computational demand of DEM limits its use with machine learning. By modelling the terrain and wheels as rigid, simulations can run significantly faster than real-time, making them a rich source of data. Through this approach, we develop a model for path planning that, given a local height map with directional dependency, predicts measures related to locomotion, energy consumption, and acceleration. When a path has been established, the problem of tracking control takes over. We tackle autonomous vehicles by learning control with perception using deep reinforcement learning. To understand the gap between simulation and reality, we transfer control policies to the physical counterpart and study discrepancies through extensive field tests.

Although most concepts in this thesis find applications in, e.g., mining and construction, we pay special attention to forestry.

Chapter 2

Background: Navigation in foresty

Within the present scope, navigating a forest terrain with a heavy vehicle is concerned with safe, efficient, and environmentally aware traversal to reach some destination. To reach the desired location, we need an overall plan of where to drive, a means of revising the original plan, and a control strategy that realizes the plan and adapts to unexpected changes. In the proposed setting, a *global planner* is responsible for the overall plan. It is used offline and considers the entire area of interest with help of maps to suggest routes leading to the final location. A *local planner* operates at a smaller scale, runs in real-time, and can revise the plan while taking the vehicle's current state into account. The local planner relies on sensors and utilizes direct information for localization, velocity, and perception to interpret the surroundings but does not necessarily provide a control strategy. In some cases, it is convenient to merge the purpose of the local planner into a *tracking controller*. Such a controller has the potential to plan, perceive, and control at a higher level than the local but below the global planner. In certain aspects, the global plan is a prerequisite for the tracking controller. If the global planner suggests kinematically infeasible paths and ignores the vehicle-terrain dynamics, the tracking controller may end up in situations impossible to navigate. Other situations may be traversable but result in poor vehicle performance, which is related to soil disturbance.

To have methods for planning and control that take the vehicle dynamics into account, we first need to understand what strategies are currently being used to navigate forest environments and the difficulties these face.

2.1 Terrain and vehicles

Forest terrain varies from having firm ground, smooth surface structure, and gentle incline to weak deformable soil, high roughness, and steep slopes. The

difficulty of a forest terrain can be classified according to the combination of three categories: ground conditions, surface structure, and incline [1, 2].

Ground conditions are concerned with to what extent the ground can handle heavy traffic. They are largely affected by the root system, vegetation, soil type, and wetness. Wet conditions are associated with low bearing capacity, making the terrain less trafficable and more prone to soil disturbance.

Terrain with a high density of boulders and steep slopes is problematic for several reasons. It requires a skilled operator that perceives the surroundings and carefully plans where the running gear should go to avoid getting stuck or causing damage to the tire sidewalls. Inexperienced drivers have notably lower productivity and lack the skill or confidence to drive through areas within the vehicle's capabilities.

Forestry vehicles are designed to have high mobility, operator comfort, and ground compliance, which refers to the ability of the vehicle's suspension system to absorb shocks and maintain ground contact. These vehicles are intended to operate in variations of all three categories of terrain mentioned above. Conventional machines run on diesel engines and are all-wheel driven with hydrostatic-mechanical transmissions. They typically have one articulation joint that connects the chassis.

In Nordic cut-to-length (CTL) logging, two forestry vehicles work together as a team. A harvester cuts and processes trees into logs while simultaneously clearing a path for a forwarder. The forwarder uses what is referred to as the main extraction routes to reach logs and loads them onto the bunk, see Figure 2.1. When fully loaded with 10–20 tonnes, the forwarder returns to a landing, where a timber truck takes over. Most forwarders have eight wheels with tracked bogies, while harvesters have either the same configuration with eight wheels, or six wheels and only one bogie pair. Nonetheless, choosing an appropriate set of vehicles can be challenging and depends on the scale of the operation and the terrain.

2.2 Soil disturbance and vehicle mobility

Navigation in forest environments presents distinct challenges due to the presence of weak soils that are prone to failure due to wheeled traffic. When the vehicle load exceeds the soil's bearing capacity, the soil undergoes plastic deformations resulting from compaction and shearing. The properties of the wheel, its dynamic interplay with the soil, and the properties of the soil itself all contribute to the extent and characteristics of the deformations.

Two main factors contributing to soil strength are the soil texture, which refers to the particle shape and size distribution, and moisture content. Fine and medium-textured soils typically have low bulk densities and are the most sensitive to changes in moisture content [3]. A high moisture content

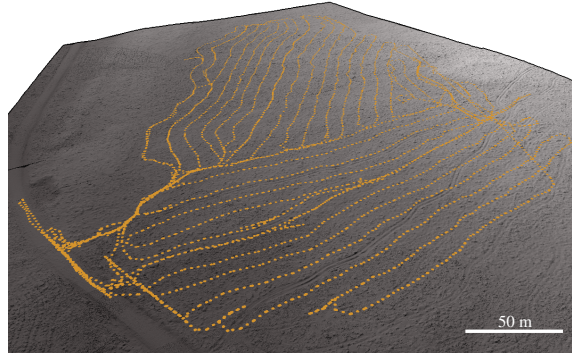


Figure 2.1: Point cloud representation of harvest area and harvester trajectories resulting from final felling. The tracks in the terrain surface indicate the layout of the main extraction routes. Data used with permission from Komatsu Forest AB.

is related to low shear strength, while soils with low bulk densities are susceptible to compaction.

Soil disturbance leading to compaction is of high concern in forestry. It reduces the porosity and pore connectivity and is not always visible, making it difficult to assess. Another concern is rutting, which occurs when normal and shear forces displace the soil to create visible tracks. It can occur as a consequence of either soil compaction or shearing but most frequently as a combination. Both rutting and compaction negatively impact the forest ecosystems [4] and affect vehicle performance.

Mobility analysis expresses vehicle performance using metrics, such as terrain resistance, drawbar pull, wheel slip, and tractive effort. Physical experiments on natural soils for mobility analysis are problematic due to the spatial variability and sensitivity to weather. Plenty of research has been dedicated to experiments in soil bins with simplified rigs of single wheels. Although these provide valuable findings, the approach is not practical to test new designs on a variety of relevant soils. The mobility metrics depend on the normal and shear-stress relationships that occur at the vehicle-terrain interface and are far from trivial to predict [5].

To fully comprehend the relationship between wheels and soil regarding mobility and soil disturbance, it is necessary to analyze the interaction through the underlying physics [6]. This involves studying the normal and shear stresses at the wheel-soil interface under dynamic circumstances. As the wheel interacts with the soil, it gives rise to a network of forces between individual soil particles that extend far below the surface. The resulting subsoil stresses play a critical role in explaining soil compaction [7] and rut formation, and physics formulations based on first principles provide a powerful means to study them.

2.3 Operational planning

Global planning is a central piece of operational planning in forestry concerned with designing a network of extraction routes. Since the extraction routes are subject to repeated loaded and unloaded passes of a forwarder, with a loaded weight normally in the range of 20–40 tonnes, their layout affects productivity and environmental impact.

Due to the effect of water content on soil-bearing capacity, the best practice is to completely avoid wet areas when possible. In cases where harvest operations require traffic through sensitive areas, the strategy is to wait until conditions are dry or when the ground is frozen. However, the continual demand for fresh wood leads forestry vehicles to operate under wet ground conditions. Soil disturbance can be mitigated by reducing soil pressure through vehicle design, e.g., larger tyres and bogie tracks, or by reinforcing the ground with logging residue, called brush mats [8]. Brush mats distribute the vehicle weight over a larger area but are limited in that the effects deteriorate over multiple passages and mainly protect the topsoil [9]. Another frequently indicated measure to mitigate soil disturbance is to improve planning for harvesting operations [8].

In recent years, network design has transitioned from in situ planning with GIS data towards more digital solutions. State-of-the-art methods use optimization techniques that rely on digital elevation models of the terrain, depth-to-water maps, and volume stand density [10].

Despite new efforts, current methods for planning extraction routes do not account for vehicle kinematics or dynamics. This limits the planning procedure in terms of selecting a suitable vehicle, considerations of fuel consumption, time requirements, and the potential to avoid areas that are difficult to traverse or may damage the vehicle.

2.4 Autonomous forwarders

Advancing vehicle design is motivated by the need to address climate change due to the emission of greenhouse gases [11] and the effects of soil disturbance due to harvest operations [4]. To allow for drastic changes, vehicles need to operate with partial or no human supervision [12]. Both cases enable designs without a cabin to reduce the vehicle weight and mitigate soil disturbance, as well as a control strategy that excludes operator comfort. In addition to aspects of comfort, harvest operations are expensive, hazardous, and demand that operators work in remote areas, which further motivates the inclusion of autonomy. Autonomous vehicles in forestry are mentioned in a multitude of recent visionary work and review papers [13, 12, 14] but have since not been properly addressed in the literature.

A gateway towards autonomous vehicles in CTL logging is to remove the

operator from the forwarder. An operator working remotely has the possibility of controlling several vehicles, granted that some subtasks are handled without human supervision. It has recently been shown that autonomous loaded and unloaded traversal of the forwarder can reduce operation costs by up to 10 percent [15]. However, the remote and cluttered nature of forest environments curbs the ease of practical application.

2.4.1 Localization and mapping

Localization and mapping are two complementary processes that are essential for autonomous navigation. While mapping creates a representation of the environment, localization determines the position and orientation of a robot within that environment. Given the current availability of terrain maps and the use of global navigation satellite systems (GNSS) on mechanized machines, global mapping and localization with sufficient precision is generally not a problem in modern forestry. However, robot applications that require local planning rely on local mapping and high-precision localization. High-precision localization must be able to cope with GNSS signals that are often occluded by the forest canopy. This has yet to exceed the experimental level in forestry [16].

A promising approach to achieve high-precision localization is through simultaneous localization and mapping (SLAM). SLAM positions the vehicle in a map of the environment that is constructed and updated online with help of onboard sensors. Several works have addressed SLAM in forest environments [17], but remain an active area of research, in particular, concerning 3D localization [18].

The challenges of SLAM can be greatly simplified in the presence of 3D maps of the environment generated in advance. Preprocessed maps can be achieved either from data collected by the harvester or from aerial laser scans. For instance, aerial scans of the entire Swedish country are publicly available and updated regularly with increasing detail for each occasion. With detailed maps, the problem reduces to localization, considering how the environment has changed since the map was generated and detecting moving objects. Localization in a known map of the environment is commonly addressed for urban vehicles with the motivation to handle high speeds with low-cost sensors for geo-spatial positioning. In forests, the issue rather resides in a weak satellite signal and the unstructured environment, which complicates feature matching between the global map and onboard sensors for vision or lidar. Recent work presents a method that uses lidar data collected from a harvester which is processed offline to create a Delaunay triangularization of points from tree trunks as a global map [19]. While online, the procedure computes a similar triangularization from a local point cloud which is matched with a subset of the global triangularization. The process then uses the match to estimate a rigid body transformation of the

vehicle. They show a standard deviation of 12 cm in location and real-time performance at reasonable speeds for a forwarder. Fusion of other sensors such as a camera can improve accuracy while including GNSS data beyond the initial state estimate can reduce computational cost. The method they present and similar contributions are of key importance for autonomous vehicles in forestry.

2.4.2 Tracking control

Tracking control relies on high-precision localization and mapping and refers to finding a control strategy that realizes a tracking objective. In the context of autonomous forwarders, the goal is to track the extraction routes devised during operational planning. As the actual extraction trails may deviate from the operational planning, any alterations are communicated within the harvester-forwarder team. From a global perspective, the harvester's localization enables the tracking of its whereabouts and the digital marking of piles of logs with coordinates. This information allows for the specification of the final forwarding route as either a geometric path or a sequence of vehicle poses, including position and heading.

Control for path tracking is restrictive in the sense that there is a predefined notion of where the vehicle should drive. Pose tracking allows more freedom as any path is feasible as long as the control strategy reaches the pose, but it may also come with higher demands on the controller in terms of perception and planning.

Path tracking control for forwarders was last explicitly addressed more than a decade ago [20]. Since then, technology concerning sensors, machine learning, and computational power has advanced rapidly, implying room for more sophisticated solutions. In recent years, new vehicle designs are emerging that aim to address ground compliance, mobility, and gentle soil interaction, yet with high productivity, e.g., the Xt28 forwarder and the Komatsu Centipede [21]. Novel designs tend to be complicated for controller design because they utilize active control more than clever mechanical design. With new designs, there is reason to believe that current control methods can be improved to help reach the full potential of the vehicles.

In the context of tracking, it has been proposed that the modules of a complete controller should address sensing and perception, planning, and control [22], taking into account both short and long-term goals. It remains a challenge to develop controllers that satisfy these criteria and adapt to forest environments while being fast enough to run online.

Chapter 3

Research focus and aim

This thesis aims to utilize physics simulations to contribute to areas of forest navigation where current knowledge is insufficient or methods for planning and control can be improved. Based on the present state-of-the-art, we have identified four research challenges that constitute the ground for our scientific contributions.

To improve the present understanding of soil disturbance and design high-mobility vehicles that can navigate weak soil, reliable simulation methods are a key component. Through the question of reliability, we identified a knowledge gap concerning the simulation of weak soil.

Paper I It is currently unclear if state-of-the-art physics simulations can capture the complex wheel-soil interaction, including the large deformations that occur in natural soil under heavy traffic.

An advantage of simulations is the ease of data collection with the intended use of machine learning to train complex function approximators. Data collection not only applies to the amount of data but also to situations that are dangerous to explore in the real world. Related to machine learning, we improved the way to train predictor models and design controllers.

Paper II Current methods for global planning in forestry ignore vehicle dynamics and do not utilize the increasingly available high-resolution topography data.

Paper III Current control methods are insufficient to utilize the full potential of novel forestry vehicle designs and make the transition from human operators to autonomy.

The consequence of simulations is that any control strategy must be evaluated in reality to ensure that it reflects real-world performance. It is not obvious to what extent the simulation-based control methods are of practical use in the real world. Therefore, we addressed the following challenge.

Paper IV Current research does not agree on the simulation-to-reality transfer capabilities concerning heavy vehicles with complex drivelines and slow actuation.

3.1 Key assumptions and consequences

The simulations in this thesis model the wheels as rigid and the terrain interaction with the ground using friction, compliance, Young’s modulus, restitution, and damping. While these parameters simulate some of the characteristics between flexible tires and the ground, they merely idealize the interaction of a real tire.

The use of a flexible tire model would introduce additional simulation parameters that require calibration. If the calibration is not accurate, there is a risk that the tire model introduces other and potentially larger errors than that of a rigid wheel model. Although tire models are crucial for vehicle dynamics simulations at high speeds on asphalt [23, 24], the importance is not as clear for heavy vehicles that move slowly over rough terrain.

Treating both wheels and terrain as nearly rigid (Papers II–IV) results in models that cannot generalize to soft and plastically deforming terrain for traversability prediction or control. Rigid wheels on rigid terrain cannot capture the increased tire flex on the lower side that occurs due to the shift in vehicle weight distribution when driving on a side slope. Models that use rigid wheels may thus overestimate traversability in such situations. A similar reasoning applies to lateral sway due to the flexing of pneumatic tires. Nonetheless, the methodology concerning Papers II–IV applies equally well to a simulation with a flexible tire model. Analysing the consequences compared to the current wheels is considered interesting future work.

When considering deformable terrain, the options for rigid tires range from empirical models to physics-based finite element models. The latter option is notoriously computationally expensive, while there is no obvious choice among other models concerning rough terrain [25]. In Paper I, a more sophisticated tire model would affect the topsoil stresses. In contrast, tire inflation pressure has little to no effect on subsoil stresses, where wheel load dominates [7].

The choice of forestry vehicle in Papers III–IV warrants further motivation. Instead of using a conventional forwarder, which has a hydrostatic mechanical transmission and bogies, we selected the Xt28 forwarder with fully hydrostatic transmission and active suspensions with wheels. Two main reasons influenced the decision: the unavailability of expensive forestry vehicles for experimental purposes and the Xt28 forwarder’s increased controllability compared to conventional forwarders. The Xt28’s six controllable suspensions allow for a higher degree of control, making it more interesting from a control perspective. Additionally, active suspensions play a more prominent role in novel forestry vehicle designs, such as the Komatsu *Centipede* [26].

A final consideration, which applies to Papers II–IV, is the absence of trees. Future research on obstacle avoidance and route planning will require additional effort to account for trees.

Chapter 4

Tools: Simulation methods

Although the work in this thesis does not deal with developing numerical methods, it is fruitful to have basic knowledge of the techniques we use to simulate vehicles and terrain.

A vehicle essentially consists of multiple bodies interconnected with joints, called a multibody system. The motion of a multibody system under influence of forces is described by its dynamics. Vehicle dynamics is typically complicated due to driving forces in actuated joints and contacts with friction. As a consequence, the equations of motion cannot be solved analytically, and thus need numerical treatment.

Simulation methods for soil-wheel interaction can be categorized as semi-empirical or physics-based [25]. Semi-empirical methods combine models based on experimental results with physics. The hybrid approach generally reduces computational effort at the cost of fidelity. High-fidelity models are physics-based that rely on physical principles. The finite element method can represent both the terrain and tire or be combined with soil represented using the discrete element method [27]. These methods come with a high computational cost, which limits their practical use with machine learning.

In the present work, we simulate the vehicle through computational multibody dynamics but treat the terrain in two different ways. Soil is modelled using the discrete element method under conditions to produce highly realistic interactions with rigid tires. In machine learning applications, computational speed is key to producing the amount of data necessary to train models. To reduce computational effort, we treat both the terrain and the tires as rigid bodies. This results in models that are only expected to perform well on terrain with relatively hard surfaces and tires with high inflation pressure.

4.1 Rigid multibody dynamics

The multibody representation of a vehicle that interacts with a rigid terrain is complicated for several reasons. Due to the reconfigurable nature of dynamical systems with intermittent contacts, generalized coordinates is not a standard approach [28]. Instead, the state may be expressed in descriptor form, which uses all six degrees of freedom for all bodies, despite the presence of kinematic constraints. As a consequence, constraint forces need to be computed explicitly, which transforms the equations of motion from a system of coupled ordinary differential equations to a set of differentiable algebraic equations (DAE). Unfortunately, DAEs are generally more difficult to solve.

Another challenge is that simulations should be fast, accurate, and numerically stable. When simulating a vehicle over a long time horizon, one essential requirement is global stability, which ensures that simulated trajectories do not drift away from the physical ones. Global stability implies the shadowing property [29], meaning that the simulated trajectory intersects the physical one at every time step and prevents model errors from accumulating over time. Variational integrators exhibit the shadowing property and are symplectic, a key property to allow for large time steps but ensure global stability [30]. The stability comes from the variational nature from which these integrators are derived — by discretizing the action principle directly [31]. Variational integrators preserve geometrical symmetries and invariants and generally exhibit qualitative physics under reasonable computational cost [32].

An effect of large time steps is that contacts, joint limits, and friction require (semi-)implicit time integration and become non-smooth. This means that velocities may change instantaneously according to some contact law. The option is to use an adaptive time step with the caveat that the dynamics might end up in situations that require small time steps. Therefore, a fixed time step is preferred if we want to know beforehand how long the simulation will take. The non-smooth behaviour can be imposed as complementary and inequality constraints together with the DAEs to yield what is known as a non-linear complementary problem.

So far, we have covered the case when the terrain is treated as a static rigid body. To represent soils requires a different method, one capable of dealing with large-scale simulations.

4.2 Discrete element method

While multibody dynamics is concerned with relatively few bodies with complex interactions, the discrete element method (DEM) [33] specializes in a large number of particles with contacts. DEM is a well-established method to simulate granular materials, e.g., soils. In DEM, the soil is treated as a collection of individual particles, often of simplified shape with sizes much

larger than the actual particles. Adding rolling resistance can mimic the behaviour of more complicated shapes, but lacks the interlocking behaviour of the actual particles. Sizes can only be increased to the length scales of the interaction tool or the deformations of interest. In full 3D, the result is often hundreds of thousands or millions of particles, and with these numbers comes the issue of computational speed.

Compared to smooth DEM, the non-smooth version lets us get away with a significantly larger time step with the caveat that each step is heavier computationally. The choice between a smooth, non-smooth, and semi-smooth approach is problem dependent. Moreover, the choice may be affected by the possibility to form a unified framework together with multibody systems.

4.3 Numerical integration scheme

Through this thesis, we rely on a numerical integration scheme based on the SPOOK stepper [30] as implemented in AGX Dynamics [34]. SPOOK is a first-order accurate integrator derived from discrete variational principles. The method uses a semi-implicit discretization and introduces regularization parameters with physical interpretation.

The resulting numerical scheme is formulated as a mixed complementary problem, which arises because the equations of motions are constrained through joints, motors, and frictional contacts. Because both the multibody system and the DEM are expressed with non-smooth dynamics we can avoid the hassle of co-simulation, where two systems are solved separately and mapped together to a final solution. However, for the applications in this thesis, the vehicle and its bodies require high precision and are therefore solved with a direct solver. The particle system on the other hand is computationally expensive and better suited to an iterative solver at the expense of accuracy. Putting the pieces together yields an efficient hybrid direct-iterative solver that is guaranteed to be well-conditioned for each step.

Chapter 5

Method: Learning to control

The field of vehicle control is exceedingly large and diverse, inarguably without a single method acting as a silver bullet. To sort through the myriad of control algorithms, a good place to start is to think about the characteristics of the governing system dynamics, commonly expressed through the state equation

$$\dot{x}(t) = f(t, x(t), u(t)), \quad (5.1)$$

where the dynamics f is a model of reality that predicts how the state variables $x(t)$ change depending on the control input $u(t)$. Decisive consideration should be related to the dynamics and whether we have sufficient knowledge to express it mathematically based on first principles. If we can formulate a reliable model of the system dynamics, model-based control methods, e.g., sliding mode control, adaptive control, and model predictive control (MPC), will always be preferred over data-driven approaches. However, when the proposed model is inaccurate, state prediction errors may accumulate over time, causing the state predictions to drift away from reality. This puts a limit on the planning horizon.

In the context of vehicle control in rough terrain, an analytical expression for the dynamics is not straightforward. The driveline, actuation, and intermittent ground contacts with stick-slip friction produce a non-smooth and strongly non-linear system. A derivation of the dynamics involves simplifications, assumptions, and heuristics but has nonetheless been successfully applied in closely related problems of legged locomotion with MPC [35]. Since MPC requires real-time optimization, the controller is limited to local planning. A further limitation relates to the unreliability in dynamics and noisy state variables, which affects the controller's robustness [36]. When the underlying dynamics are unknown, unpredictable, or expensive to evaluate, there is reason to explore data-driven approaches.

5.1 Data-driven methods

The common ground for data-driven control methods is to build approximate models to solve control problems based on data. The data can be collected in the real world, in simulation, or by combining the two.

In applications of rough terrain vehicles, real-world data is not easily accessed. Even when forestry vehicles are equipped with adequate sensors, the data is generally noisy and limited to safe and well-tested driving scenarios. The issue of safety prevents exploring the response of controls that may result in dangerous situations but have the potential to improve or infer valuable information regarding the control policy. Limited exploration capabilities also have implications for building a model of the system dynamics from data. Simulations of multibody dynamics alleviate the issues of real-world data collection in an effective way to provide machine learning methods with a sufficient amount of data to produce powerful function approximators [37]. However, as no simulation is perfect, it is not obvious to what extent behaviour learnt in simulation transfers to reality.

The remainder of this chapter presents a concise review of data-driven control methods, which can be categorized as model-based or model-free. Extra emphasis is put on simulation data and the model-free approach, in particular for control problems of ground vehicles on rough terrain.

5.1.1 Model-based

Model-based methods, e.g., system identification and neural network control, approximate the system dynamics based on data and use the obtained model for control optimization. This method finds practical use in situations where the dynamics of a specific subsystem are complicated, unknown, or expensive to evaluate, but ground truth data is readily available [38, 39].

Contrary to the name, model-based reinforcement learning (RL) is viewed as a hybrid method as it iterates between learning a dynamics model and searching for an optimal control policy. While model-based RL is efficient and likely to become a widely adopted method for complex robotic applications, it is not as mature as its model-free counterpart [40, 41].

5.1.2 Model-free

If the system dynamics is very complex and unpredictable, then it intuitively makes more sense to try and learn how to act in the environment than to learn about the environment itself. This approach closely resembles how living organisms learn without explicitly considering the laws of physics.

In the RL setting, the environment is the world the controller lives in. Formally, the environment is described as a Markov decision process (MDP) expressed as a tuple $(\mathcal{S}, \mathcal{A}, p, \gamma, r)$, where $\mathcal{S}$ is the state space and $\mathcal{A}$ is the

action space. A controller in state $s_t \in \mathcal{S}$ takes an action $a_t \in \mathcal{A}$, and the environment responds with a transition to a new state s_{t+1} according to the state-transition probability $p(s_{t+1}|s_t, a_t)$ and a reward $r(s_t, a_t)$. The scalar reward reflects the quality of the previous action, while the discount parameter γ determines the importance of short-term compared to long-term rewards. Because the consequence of an action only depends on the current state, the process satisfies the Markov property. It means that the current state is sufficient to predict the future and all history can be disregarded, i.e., $p(s_{t+1}|s_t, a_t, \dots, s_0, a_0) = p(s_{t+1}|s_t, a_t)$.

Unfortunately, real-world robotic applications are rarely Markovian due to imperfect sensors and the inability to observe the full state. Problems with partial observability are commonly modelled as a partially observable MDP (POMDP). Although POMDPs are often sufficient for robot control, partial observability typically affects optimality as it presents itself as noise and randomness in the state-transition function.

Without a model of the state-action transitions, control problems within the paradigm of model-free RL use trial and error to find a policy that maximizes the sum of future rewards [42]. The policy is a strategy the controller uses to decide how to act, and an RL algorithm determines how to learn it. This algorithm relies on an interactive process between the controller and the environment to collect data, which it uses for policy updates. Based on the policy used for data collection, RL algorithms are categorized as either on-policy or off-policy methods.

The main disadvantage of model-free RL is its sample efficiency, which is particularly true for on-policy methods, e.g., proximal policy optimization (PPO) [43]. Since these methods update the policy based only on data collected using the current policy, past experience becomes useless, making learning ineffective, especially for high-dimensional tasks. Off-policy methods address this issue by evaluating and improving a different policy than the one used to collect data. This way, they reuse previous experience and reduce sample complexity. While state-of-the-art off-policy methods, such as soft actor-critic (SAC) [44] may suffer from instability [45], they tend to outperform on-policy methods like PPO [46]. However, the greatest potential advantage of using off-policy methods is the implication to learn offline.

In offline RL, the data is collected once, possibly in the real world, by some potentially unknown policy, and that static dataset is used to train the policy. This approach is appealing because it resembles supervised learning, that a sufficiently large dataset can be used to train policies according to any objective [47]. While offline RL is a promising line of research, problems such as those related to exploration in a static dataset, may prevent it from triumphing methods that leverage simulators in an online setting [48]. Of course, the crucial requirement is that the simulators manage to capture the important characteristics of the real-world application.

5.2 The gap between simulation and reality

When we rely on simulation for learning with the intent of real-world performance, we need to be aware that the data used for control come from two different domains. Domain generalization is the ability to generalize from one domain to another, where sim-to-real is a concrete example. The special interest in sim-to-real concerning RL is due to the many complex robotics tasks solved in simulation that often do not perform at the same level in reality. The reasons stem from a wide variety of sources, such as unmodeled dynamics, randomness in the real environment, unrealistically rendered images, and a mismatch between actuator dynamics. This has led to a surge of research for methods to overcome the reality gap and achieve more efficient sim-to-real transfer [45, 49, 50].

The most obvious approach is to make the domain shift between the simulation and reality as small as possible. Realistic simulators are of great help but generally show discrepancies in visual data and rely on finding model parameters that match the true dynamics. System identification uses real-world data to find a suitable set of simulation parameters. However, factors such as temperature are rarely modelled yet may have a significant effect on actuator dynamics. Similarly, wear and tear of the equipment can cause the behaviour of the real vehicle to change over time.

Relating to the actuator dynamics, one of the most common sources of model mismatch is due to delays in the real system [45]. Latency in the actuators violates the Markov assumption and can have a serious effect on learning. The general strategy is to include the previous control signal in the observation space and extend the policy to have some kind of memory. Recurrent neural networks, such as long-term short memory (LSTM), can store information related to history but have a reputation for being unstable and slowing down learning. A similar but simpler approach, which has proven at least equally effective, is to augment the observation space with several windows dating back in time. This gives the controller the possibility to extract information relating to how and when a previously applied control will take effect, thereby restoring the process to its original Markov formulation. However, even though the simulation contains the full state, only the part that can be sensed is supplied as observation to the controller, meaning that the process is not Markov to begin with. The same is surely true for reality, which is better expressed as a POMDP with randomness and noise.

When there are uncertain modelling aspects, or we want to make the controller robust to certain variations in the domain, a common approach is to use *domain randomization*. The idea is to randomize parameters in the simulation and thereby aim to encompass the data distribution of the real-world domain. Common simulation parameters to randomize are contact friction, body's inertia and centre of mass, as well as manipulation of the simulation-rendered images. Several studies have shown the benefits

of domain randomization to produce robust controllers in the real world, ranging from object manipulation [51, 52] to legged locomotion [53], even in rough terrain [54].

Domain adaptation is another concept to improve sim-to-real transfer that mainly focuses on images and the fact that the feature space in simulation differs from that of reality [55]. Adapter networks utilize a generative adversarial network (GAN) that learns a bi-directional mapping from simulation to real images. They can be trained using either real images or simulated images with domain randomization to be completely unrestrained from real data. The controller can then be trained in simulation while observing images that go through the adapter network to look like reality. Similarly, a controller deployed in the real world can let images pass through the adapter network to produce simulation-like images [56]. Domain adaptation is a recent and promising approach that has yet to be applied to locomotion tasks in rough terrain.

The field of sim-to-real transfer is relatively new, and many aspects remain unknown despite ongoing progress. The specific factors for successful transfer vary depending on the application. However, the most prevalent cited factors are an accurate model for the actuator dynamics, modelling actuator delays, and utilizing domain randomization [45]. While prior research has focused primarily on robotic applications, such as grasping and object manipulation, locomotion tasks have nearly been limited to legged robots, with only a few addressing rough terrain [57]. Given that these robots are mostly lightweight with electric motors and fast actuation leaves a knowledge gap concerning heavy vehicles with hydraulic drivelines and slow actuators.

Chapter 6

Results: Summary of papers

Paper 1

Discrete element modelling of large soil deformations under heavy vehicles

In this paper, we answer if large deformations resulting in wheel ruts induced by heavy traffic can be captured using simulations of multibody dynamics for the vehicle and the discrete element method (DEM) to model the soil.

In the DEM models, collections of spherical particles represent soils. These particles interact through contacts according to the simulation parameters. To represent natural soils, we vary the simulation parameters and simulate the triaxial and cone penetrometer test to determine the soil bulk properties. The bulk tests showed the capability to represent a wide range of soils ranging from purely frictional to strongly cohesive. However, due to the packing procedure, the soils were dense and insusceptible to compaction. Dense frictional soils represent packed sand, while dense cohesive soils represent saturated soil with high clay content.

To study the rut formations and multipass effects, we subject a selection of soils with different characters to repeated passes by a quarter vehicle with a bogie and two wheels. The resulting rut formations, which develop prominently due to shearing, are compared to empirical models for rut depths and an experimental study. The rut depths agree with the empirical models in that weak soil and higher vehicle weight produce deeper ruts. For the multipass simulations, the majority of the rut is due to the first pass. Stronger soils are less sensitive to repeated passes. The general trend is consistent with empirical models and physical experiments.

We conclude that a soil representation through DEM can be used to study large deformations in weak soils that result in ruts.

My contributions to this paper include performing all simulation experiments, analysing the resulting data, and serving as the lead author.

Paper 2

Learning multiobjective rough terrain traversability

In this paper, we address the absence of vehicle dynamics and lack of performance-related measures in methods that analyze traversability for ground vehicles with high mobility in rough terrain.

For high-mobility vehicles, traversability as a simple go-no-go assessment is insufficient for detailed global planning. The concept is therefore extended to a multivalued function in terms of the ability to maintain forward motion, energy consumption, and acceleration. We acknowledge the increasing availability of high-resolution topography data and identify potential benefits in assessing traversability at arbitrary locations within the maps. While multibody simulations of vehicles driving on 3D representations of natural terrain can capture the vehicle dynamics, the computational demand is too high to analyze large areas within reasonable time frames.

To enhance planning capabilities in rough terrain, we propose a concept that uses a predictor model which relies on a height map and intended forward speed to predict traversability. The predictor model is trained using a supervised machine learning approach where simulation data acts as the ground truth. The simulated data consists of an articulated vehicle driving on procedurally generated terrains while sampling the local height field and the resulting traversability metrics. The results show that the model depends on the vehicle dynamics, local terrain features, and heading, which is not possible through local roughness and slope alone.

We conclude that multibody dynamics simulations can be used to train a predictor model that couples vehicle dynamics with terrain features to useful metrics related to traversability. The swift evaluation over large areas where high-resolution topography data is available provides valuable input towards vehicle design, selecting a suitable vehicle, and path planning.

I contributed to this paper as a co-author, analysing the results, and conducting simulations related to the manually driven route and feature sensitivity.

Paper 3

Control of rough terrain vehicles using deep reinforcement learning

In this paper, we explore using deep reinforcement learning to train a controller that perceives and plans to control a novel concept forwarder with active suspensions in rough terrain.

We identify the challenge of controlling vehicles to make clever use of active suspensions, wheel propulsion, and steering to navigate terrain features such as obstacles and slopes. In a forestry-related scenario, the goal of the controller is to reach a target pose through efficient and safe traversal. The target can be placed sufficiently far away to force the controller into decision-making regarding wheel placement and whether to drive over or circumvent obstructing obstacles. To make adequate decisions, we provide the controller with vision in terms of a local height field, which co-moves with the vehicle.

We train the controller in multibody dynamics simulation on increasingly challenging pose placements and terrain. To reach the intended behaviour, the reward function requires careful thought and testing through trial and error. The central challenge is to balance the possibly conflicting objectives so that the controller does not choose to ignore or overly prioritize certain terms. For instance, to align the vehicle heading with the pose, but to some extent discard the final position.

With a dense and well-balanced reward, the training resulted in a controller with all the necessary skills to navigate rough terrain. The results show intelligent use of the active suspensions to handle steep slopes in any direction and maintain ground contact when dealing with obstacles. The controller uses the local height map for perception and decides to drive around impassable obstacles. When obstacles appear within the vision that reveals challenging or even impossible scenarios, the value function shows a significant drop, which describes the controller’s interpretation of the quality of the state.

We conclude that deep reinforcement learning is a feasible approach to develop controllers for complex tasks that rely on vision for intelligent traversal in rough terrain. Although the control policies were tested on 3D representations of actual terrain with promising results, the question of transfer capabilities to real-world applications remains uncertain.

My contributions to this paper were to perform all the simulations, analyze the resulting data, and serve as lead author.

Paper 4

Sim-to-real transfer of active suspension control using deep reinforcement learning

In this paper, we extend the concept of Paper 3 of developing controllers using deep reinforcement learning and address the transfer capabilities of simulation-based learning to reality. The strategies for successful simulation-to-reality transfer are well-tested on lightweight robots with electric motors and fast actuation. However, less is known about heavy vehicles with hydraulic drivelines, complex coupling effects, and slow actuators.

The most widely adopted strategies for simulation-to-reality transfer are to have an accurate model of the actuator dynamics, model the actuator delays and noise in state variables, as well as to utilize domain randomization. We pay special attention to the actuator dynamics and apply system identification. For a carefully selected number of scenarios, we send control signals to the real vehicle and record the response. We then pass the same control signals to the simulation model and tune the simulation parameters that govern the actuation and terrain interaction to agree with the physical response.

We train policies in simulation with and without observation noise and delays for the control signal. To promote a smooth control signal, we allow some policies to include a penalty for changes in the control signal in the reward function. The bang-bang control, which otherwise tends to develop, is undesired due to wear and tear on the equipment, high energy consumption, and excitation of high-order dynamics that exacerbate transfer to reality. We assess the simulation-to-reality transfer of the policies by testing them on the physical vehicle, exposing the RL controller to scenarios that include driving and turning on level ground, navigating obstacle courses, and passing ramps.

The results show close alignment between simulation and reality, in particular for the policies trained with control delays and a penalty for hectic control changes. The learnt strategy to control the active suspensions shows a strong resemblance with a hand-crafted controller that has been thoroughly tested and verified to work well in practice. However, due to safety restrictions, the RL controller was not allowed to utilize the full range or speed of the suspensions and was therefore limited in mobility. Because of the complex driveline, learning was more challenging than in Paper 3, and as a consequence, there were no signs of the controller making use of the local terrain data for navigation purposes.

Our study demonstrates that an accurate model of the actuator dynamics using basic system identification and the inclusion of control delays are sufficient for successful transfer to reality. However, the complex driveline inhibited rapid learning, indicating the need for further effort to develop a controller that effectively utilizes exteroceptive information, in particular for planning. Encouragingly, the compelling results regarding transfer capabili-

ties enable us to confidently confine additional development to simulation.

In this paper, my contributions include performing all the simulations, analysing the resulting data, and serving as the lead author. I also planned and participated in the physical experiments as well as implemented the RL controller on the real machine.

Chapter 7

Conclusions

The main conclusion of this thesis is that physics simulations combined with machine learning offer a feasible and powerful means to enhance navigation strategies in rough terrain.

Regarding soil that deforms under the influence of wheeled traffic, multi-body dynamics with DEM enables analysis of physical forces around the wheel-soil interface. It is currently unclear what constitutes soil damage in forestry [4]. The ability to analyze the physical consequence of different running gear designs and soils is crucial to understand the effects of soil disturbance on the ecosystem.

The current limitation of DEM is the computational effort, which restrained us from doing the trafficability (Paper II) and RL controller training (Papers III–IV) on deformable soil. Recent progress that applies model-free RL for control on deformable terrain is a promising step forward but remains limited in terms of point contacts, blind locomotion, and level ground [58]. To make the model-free RL approach viable for such problems, progress must be made in terms of sample efficiency and computational speed. Multiscale methods, which treat the soil as a continuum until rigid bodies agitate the soil into a DEM representation, are an interesting development to speed up multibody dynamics simulations with deformable terrain. However, these currently struggle to run in real-time and lack validation with real-world data concerning wheel-soil interaction [59].

While domain randomization in model-free RL can produce robust controllers with successful zero-shot transfer to reality, it does not solve the issue of online adaption. Model-based RL methods like DayDreamer [60] address the sample inefficiency and absence of online adaption in model-free methods by learning a transition model of the environment and using a latent space representation to plan in imagination. Such methods have the potential to capitalize on real-world data while leveraging simulations where real-world interaction is unsuitable, e.g., during exploration. Making the best out of two worlds indicates that model-based methods are the way towards robust

controllers that adapt to changes in the real world.

This thesis demonstrates the potential of simulations and machine learning applied to navigation in rough terrain. Our findings provide compelling evidence that full autonomy is achievable. In addition, the methodology and models developed in this thesis are building blocks that have immediate practical applications in semi-autonomous navigation and precision planning. In summary, we see that the contributions in this thesis can aid future research in understanding soil disturbance, further enhancing route planning, and advancing autonomous ground vehicles in forestry.

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